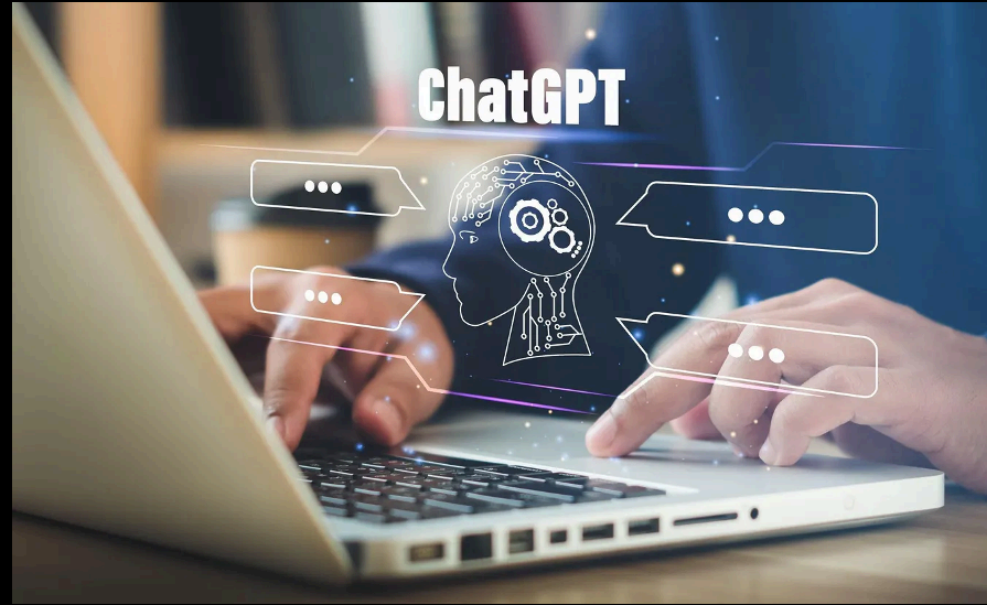




Introduction to Data Mining

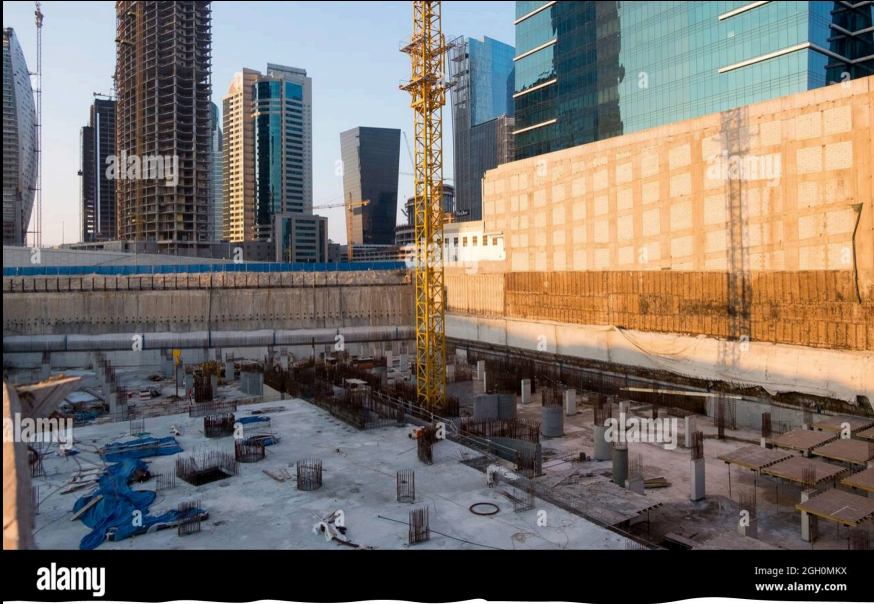


You chose IE for this ...





... but you need to start from this





Course Organization

TAs:

Ms Ra Suryeon

Mr Kim Youngjin

To be confirmed - details will be on Blackboard soon

Course Material and Announcements:

On **Blackboard**



Evaluation (tentative)

Attendance and attitude: 10%

Final exam: 40%

At the end of the course (probably on the day of the last lecture)

Individual assignments: 50%

Two “mini” practical projects

Deadlines around mid-term and late November

Discussed offline with me and/or TAs



Textbook

An Introduction to Statistical Learning (with Applications in Python)

James et al. Springer, 2023

Free online: www.statlearning.com

Covers 90% of the topics

Each chapter concluded by a *Lab* section, which is important for individual assignments (Python)



What is Data Mining?



What's the Difference?

Statistical Learning

Popular with statisticians

More focus on mathematical theory

Data Mining

Popular with database experts

More focus on data management and performance

Machine Learning

Popularized by CS (AI) experts

More focused on applied problems with real-world data

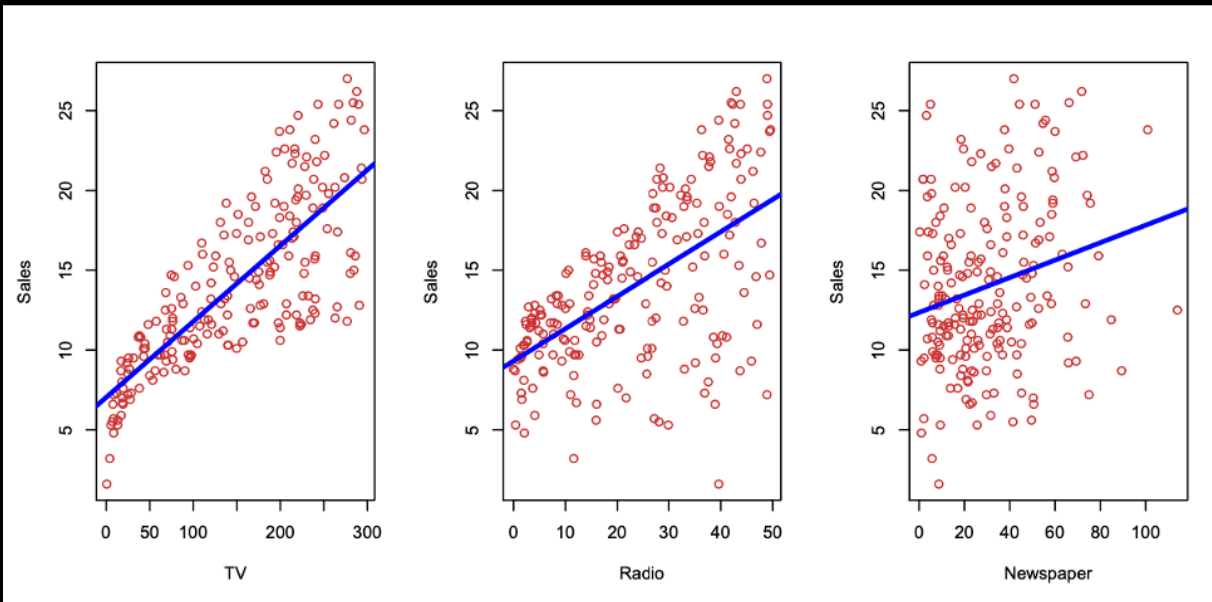
For an introductory course, they are basically the same... but don't say it loud 🤔



How Do We Increase Sales?

Sales of a product in $n = 200$ different markets for three different media

We cannot simply *increase sales*, but we can set the budget for each media, thereby (hopefully!) indirectly increase sales





Some Terminology

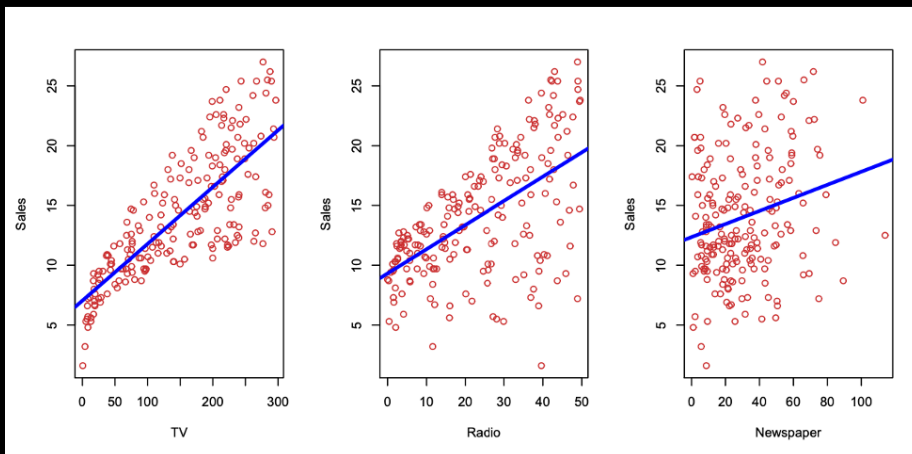
Advertising budgets on each channel are the **predictors** $X = (X_1, \dots, X_p)$

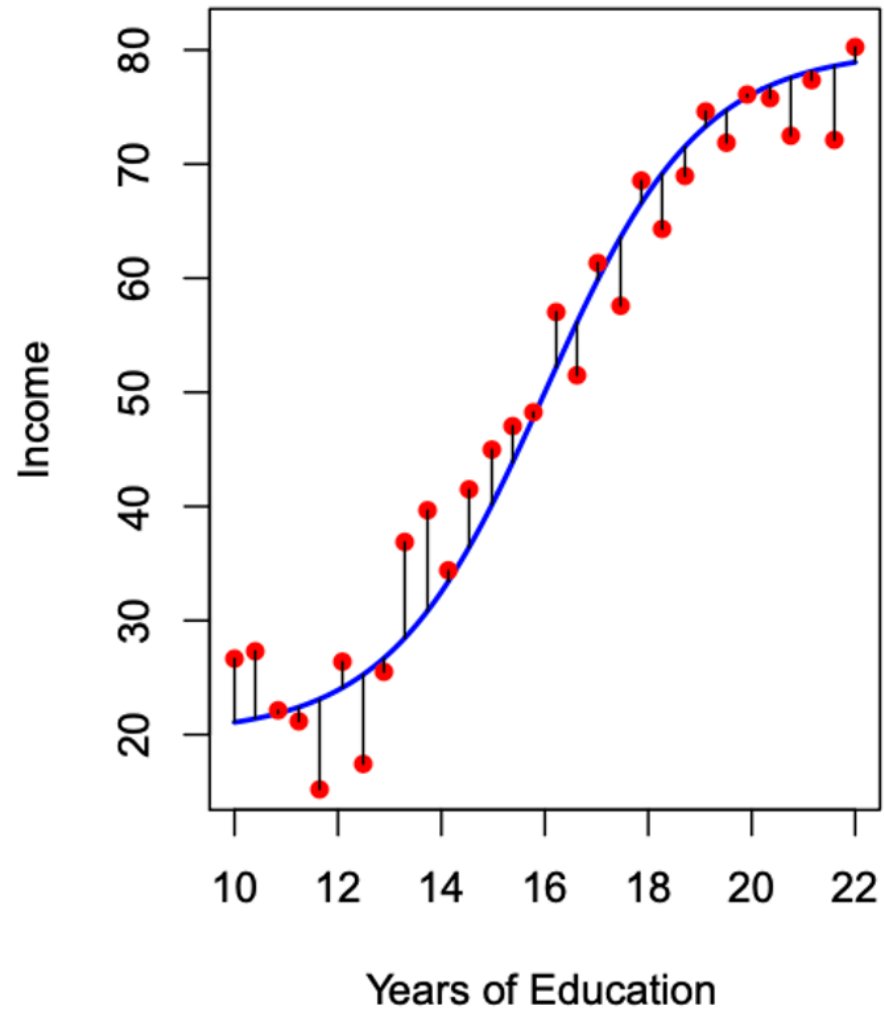
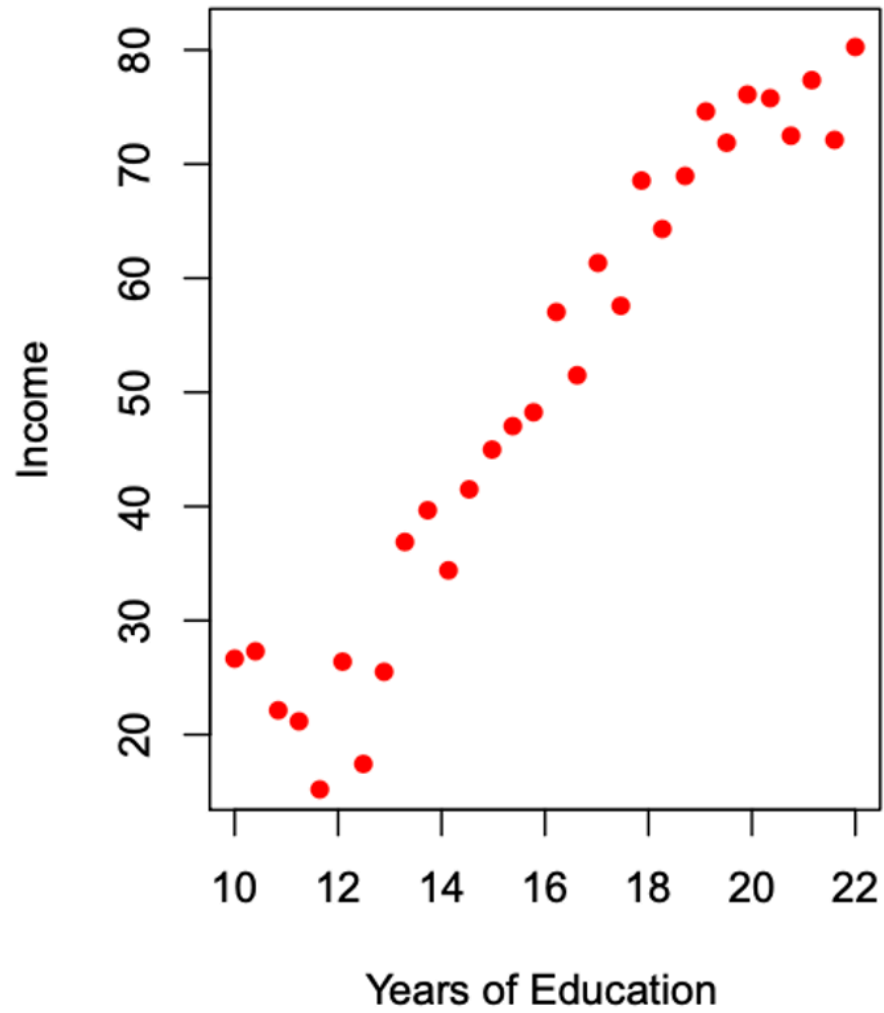
a.k.a. independent variables, features, (input) variables

$p = 3$, $X_1 = TV$, $X_2 = Radio$, $X_3 = Newspaper$

Sales is the **response** Y

a.k.a. output variable, dependent variable







What is Data Mining?

We assume that there is a relationship between Y and $X = (X_1, \dots, X_p)$ that can be written in the very general form:

$$Y = f(X) + \epsilon$$

where ϵ is a random term independent of X and with mean equal to 0

Data Mining (Statistical Learning, Machine Learning) refers to a set of approaches for deriving an estimate $\hat{f}$ of f using the available observations of X and Y .



Why We Estimate f

Prediction

Inference





Prediction

Assume we have an estimate $\hat{f}$

Often a set of input observations of X are available, but their respective output Y cannot be easily obtained

In this setting, we predict Y using:

$$\hat{Y} = \hat{f}(X)$$

Example: $X_1, \dots, X_p$ are characteristics of a patient's blood sample, Y is the risk of the patient developing an adverse reaction to a given drug



Prediction (cont.)

$$\hat{Y} = \hat{f}(X)$$

The estimate $\hat{f}$ is often treated as a **black box**

The exact form of $\hat{f}$ does not matter, as long as it yields accurate predictions $\hat{Y}$



Inference

Sometimes understanding the association between Y and $X = (X_1, \dots, X_p)$ is more important than predicting the values of Y

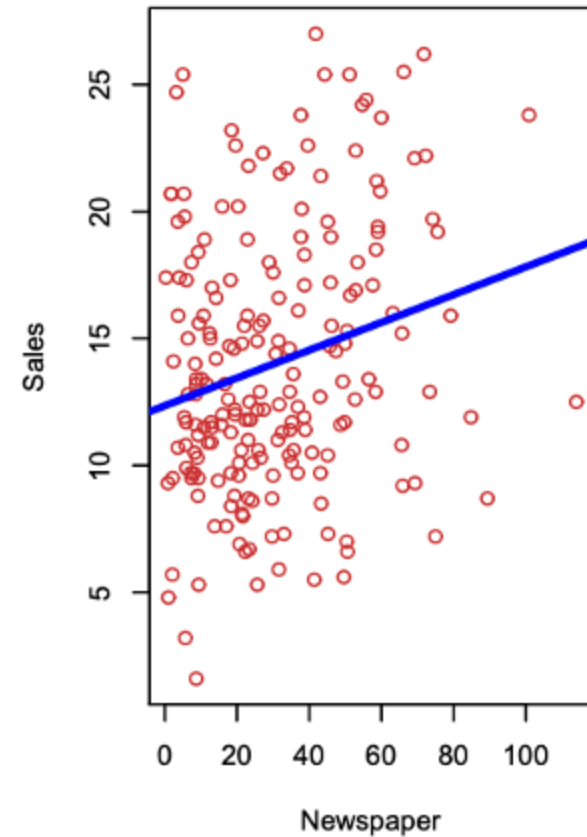
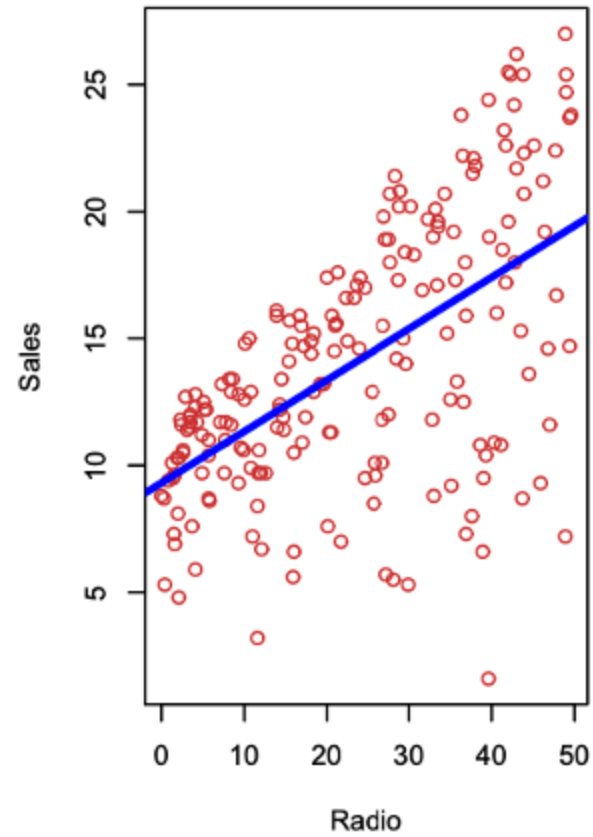
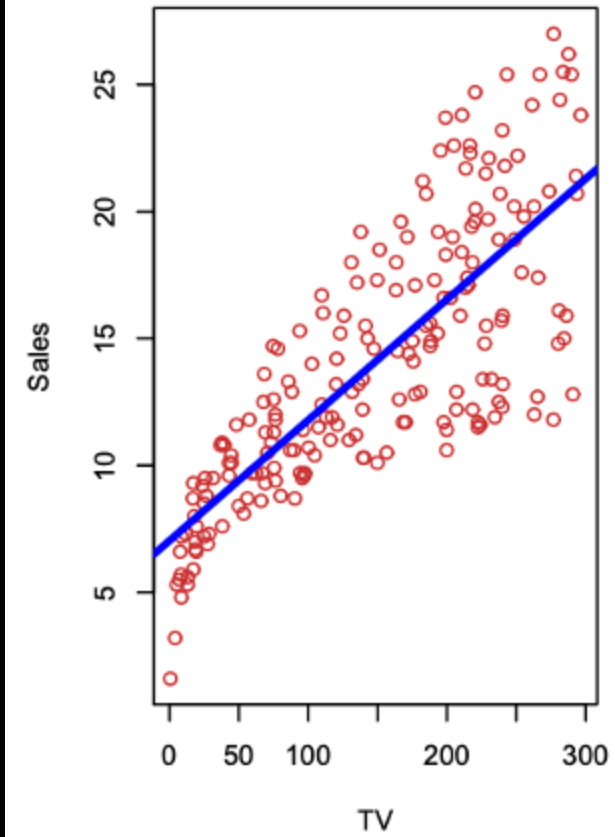
- Which predictors are associated with the response?

- What is the relation between the response and each predictor?

- Can such relation be captured by a linear equation or is it more complex?

We still wish to estimate f , but our goal is not necessarily to make predictions for Y on new observations

We cannot treat the estimate $\hat{f}$ as a black box, need to know its (exact) form





How Do We Estimate f ?

We always assume that we have observed a set of n different data points (**observations**)

x_{ij} is the value of the j -th predictor for observation i

$i = 1, 2, \dots, n$

$j = 1, 2, \dots, p$

y_i is the response value for the i -th observation

We call this the **training data**:

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, where $x_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T$



Parametric Methods

Two-step approach:

First, we make an assumption about the form of f , in which f is fully specified by a limited set of parameters

Example

f is linear in X : $f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$

To estimate f , we need to estimate only the p parameters $\beta_j, j = 0, \dots, p$

Second, we apply a procedure that uses the training data to fit (or train) the model (for linear regression, for example, we apply the *least squares* procedure)



Non-Parametric Methods

Do not make explicit assumptions on the form of f

We seek an estimate $\hat{f}$ of f close (but not too much!) to the training data points

Usually require a very large number of observations to produce an accurate estimate of f



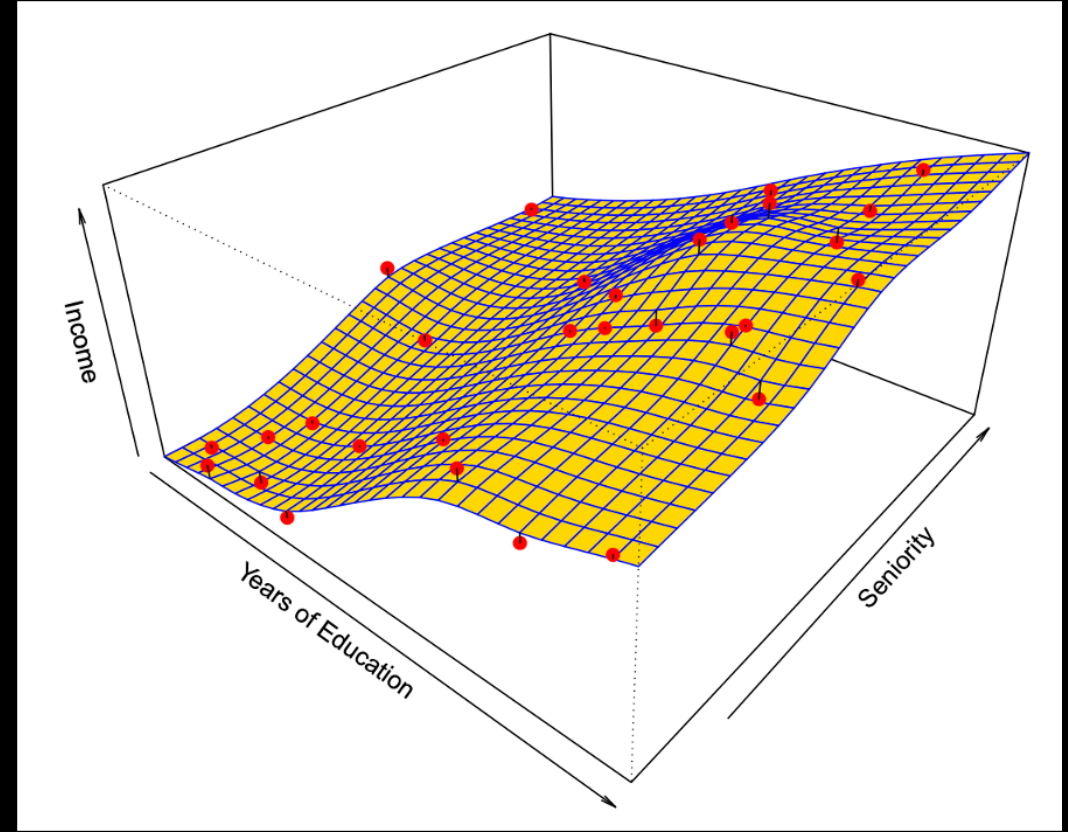
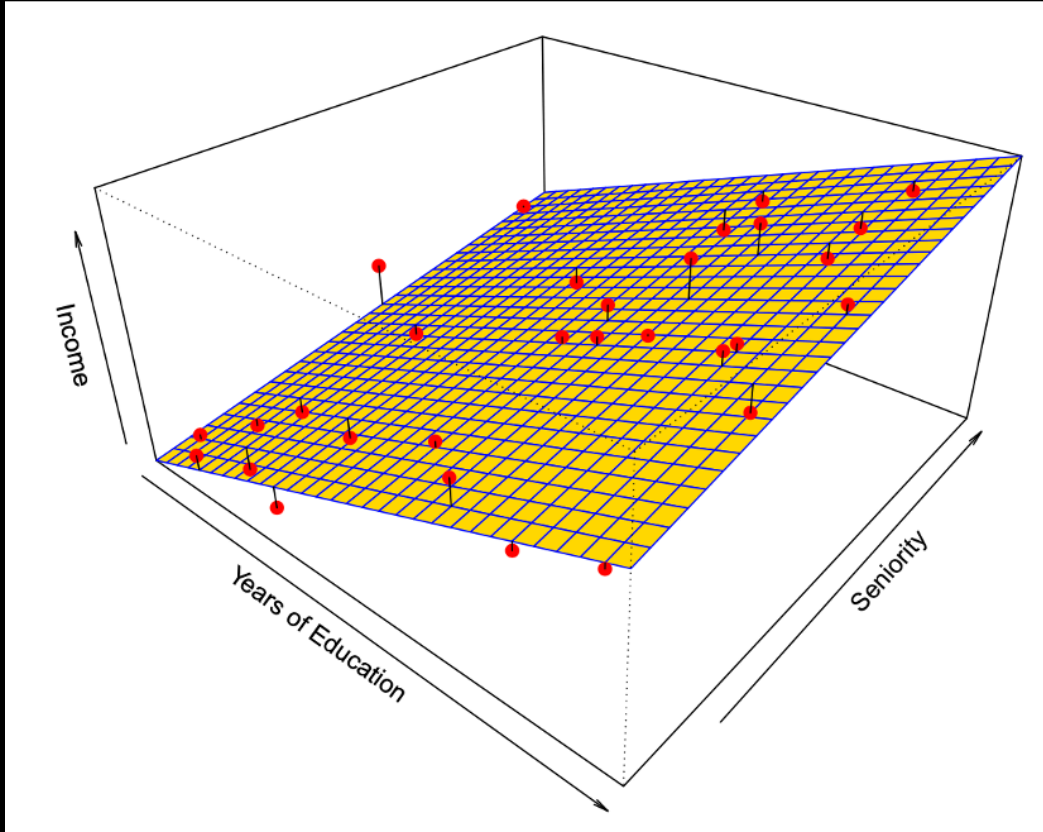
Model Flexibility

A general concept, better understood with parametric methods

In a parametric method, we need to estimate only a (usually limited) set of parameters

The model $\hat{f}$ usually does not match the true form of f
 f is hardly linear in the real world 😊

Flexible models can help by fitting many different functional forms, but they may overfit (see later)





Then, Why Choosing a Less Flexible Model?

Quality: More restrictive methods are less likely to overfit the data

Interpretability (Explainability): More restrictive (parametric) methods are usually more interpretable (for inference)

(Computational) Cost: More restrictive methods may take considerably less time to fit a model

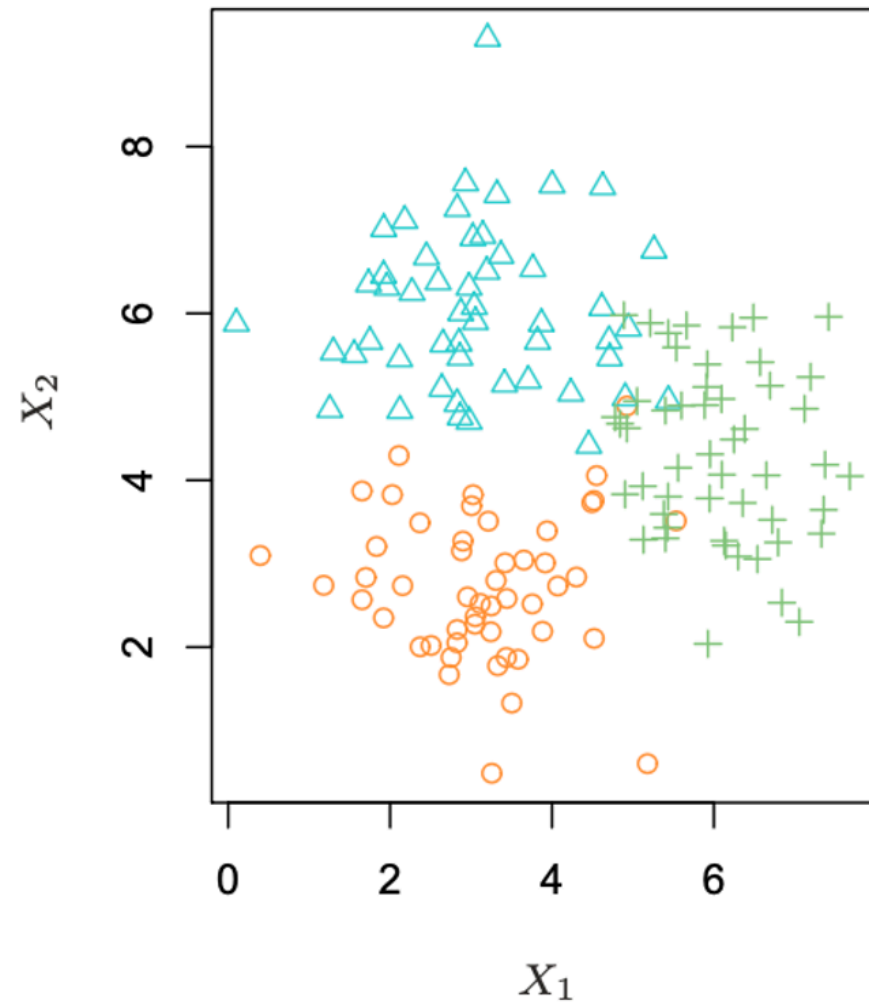
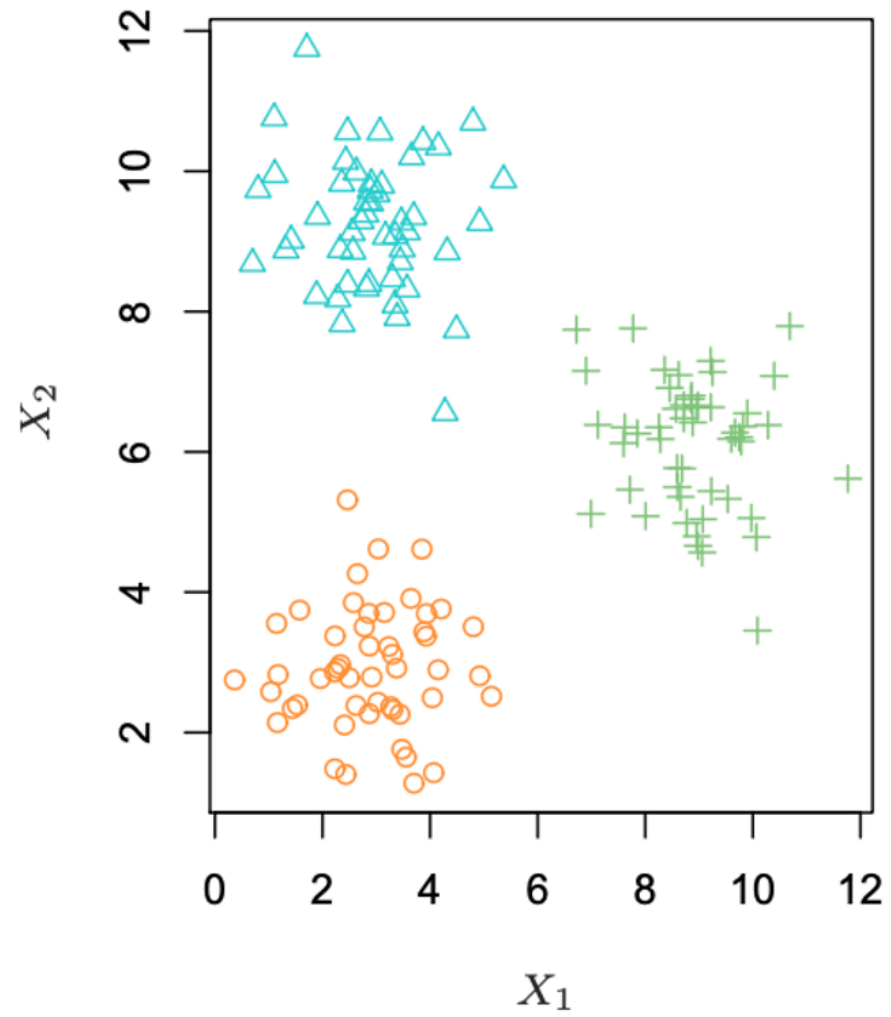


Supervised v. Unsupervised Learning

Until now we discussed the **supervised** learning domain

For each observation of the predictors X_i , we know the associated response y_i
We fit a model that relates the predictors to the response to accurately predict future observations (prediction) or to better understand the relation between the response and the predictor (inference)

In **unsupervised** learning, for each observation we have a vector of measurements X_i , but we do not have the associated response
We seek to find relationships between the variables
Clustering: group the observations in distinct groups based on similarity





Regression v. Classification

Variables can be *quantitative* or *qualitative*

Quantitative: age, height, income, prices, ...

Qualitative (categorical): marital status (married or not), brand of a product (A, B, or C), type of cancer diagnosis, ...

Regression: problems with a quantitative response Y

Classification: problems with a qualitative response Y

Linear regression is a regression method

Logistic regression is a *classification* method 😊



Quality of Model Fit (Accuracy)

To what extent the predicted response $\hat{Y}$ is close to the true response value?

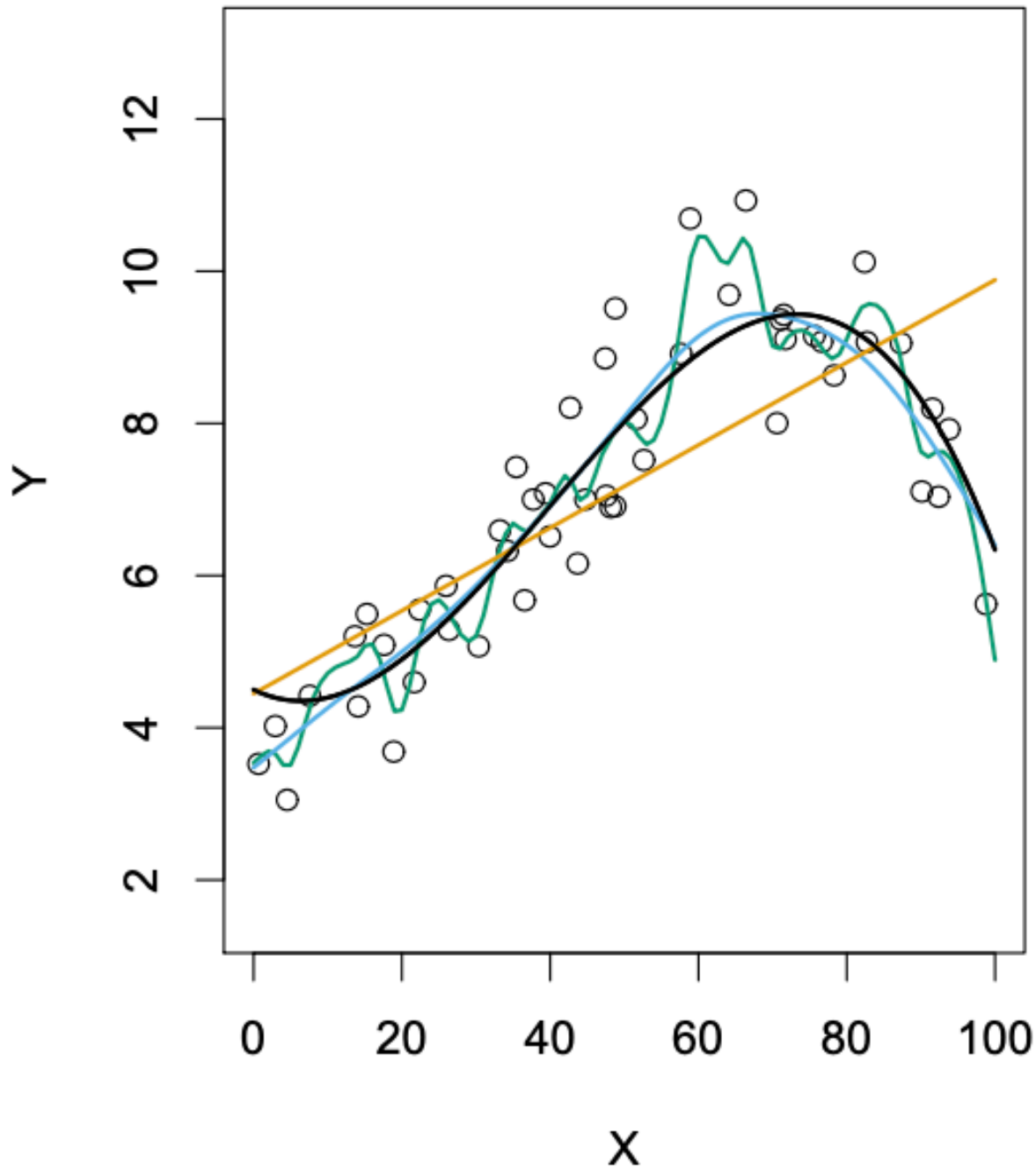
In regression, we can use the Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

This is the *training set* MSE (we can always calculate it!) ...

... but we are more interested in the error on unseen test data (*test set*):

$$Avg(y_0 - \hat{f}(x_0))^2$$



How do we reduce the training set MSE?

That's easy... use a more flexible model

But a model that minimizes the training MSE is basically *overfitting*

Learning the training data, but not able to estimate well unseen data

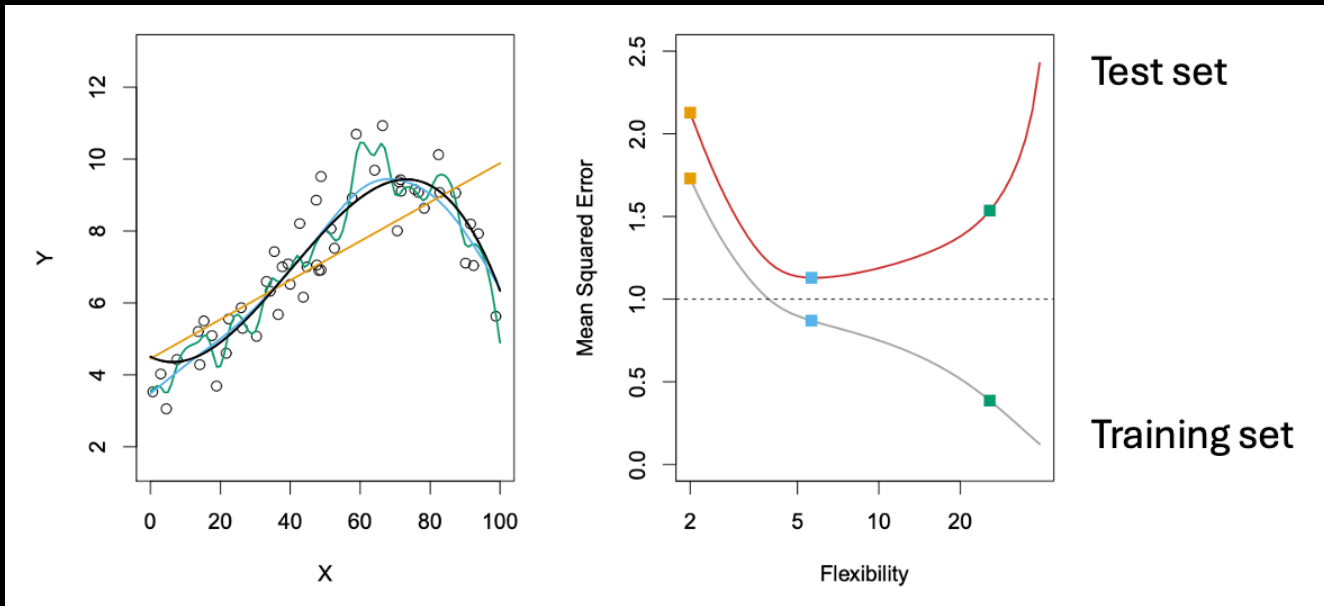


Training v. Test MSE

The training MSE constantly declines with increased model flexibility

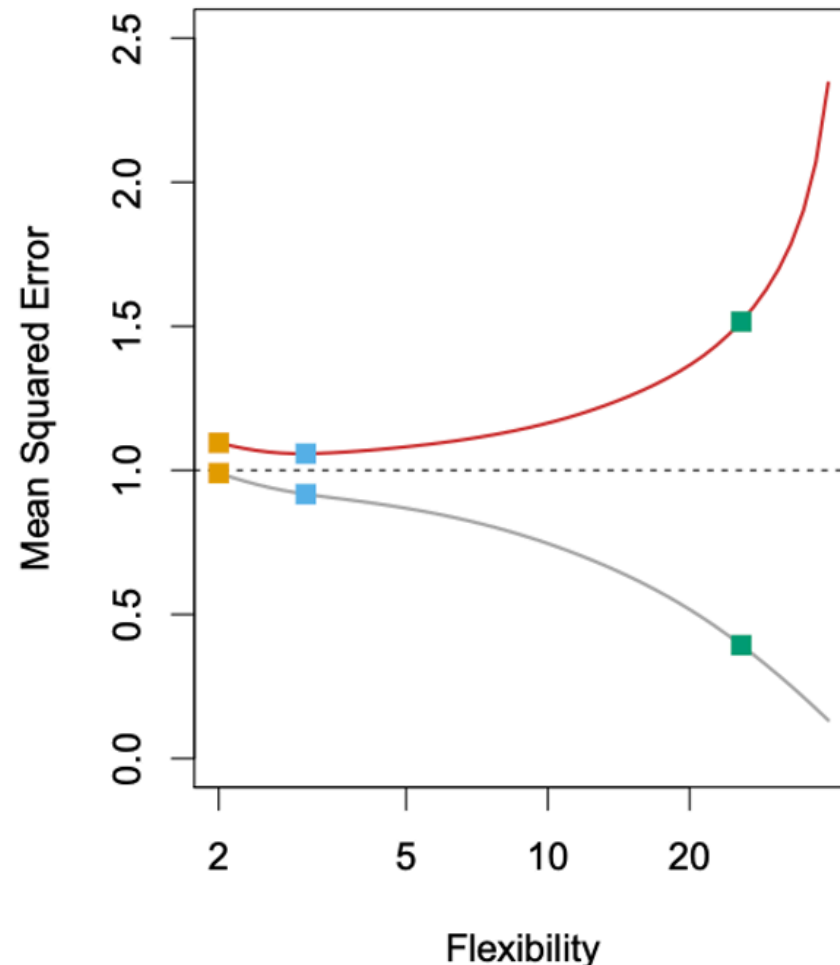
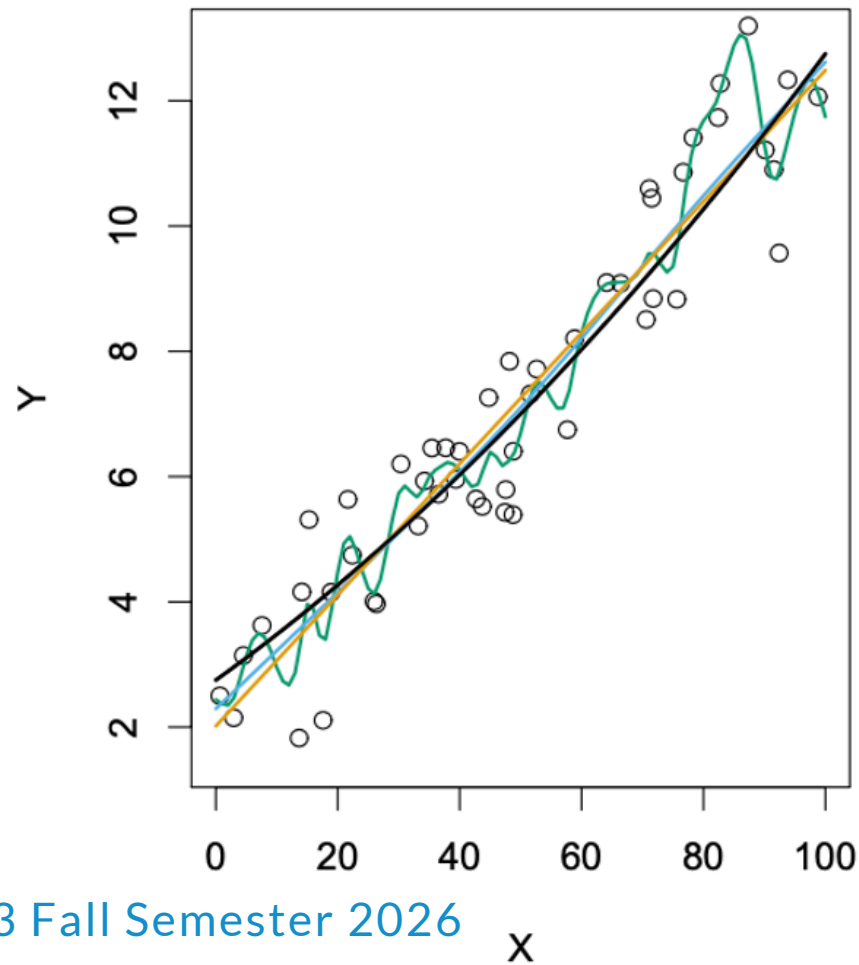
The test MSE instead shows a U-shape: it declines, but then rises again

When test and training set diverge, the model is starting to overfit the training data: that's when we should stop!





It's a General Principle!





Why the U-Shape? Bias-Variance Trade-Off

$$Y = f(X) + \epsilon.$$

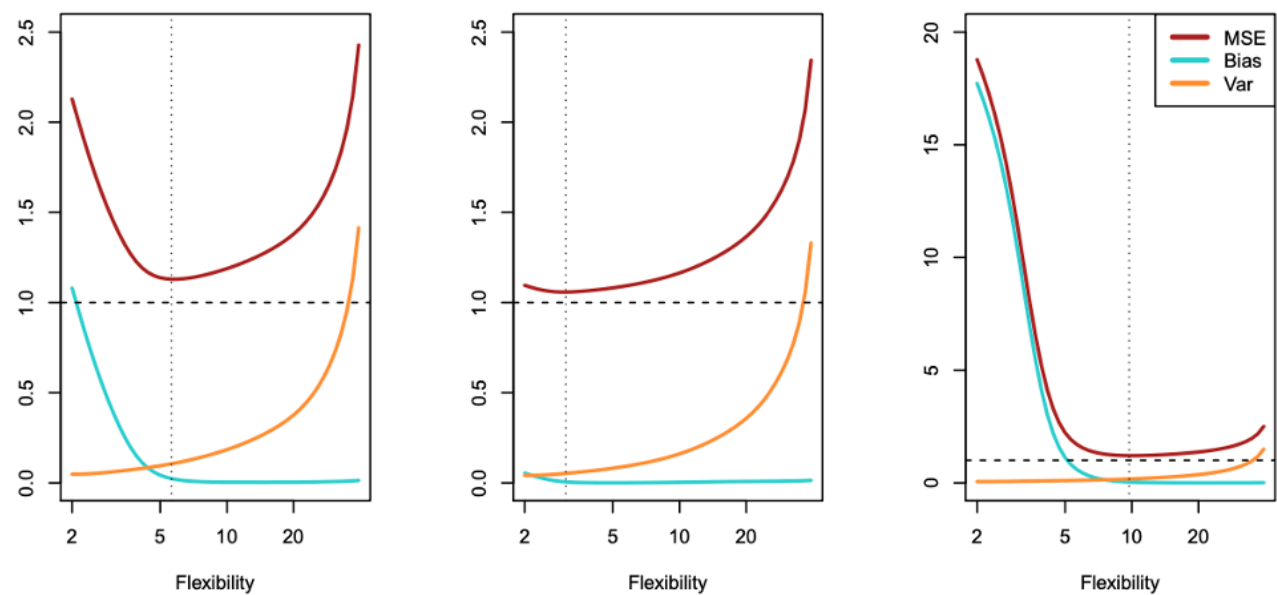
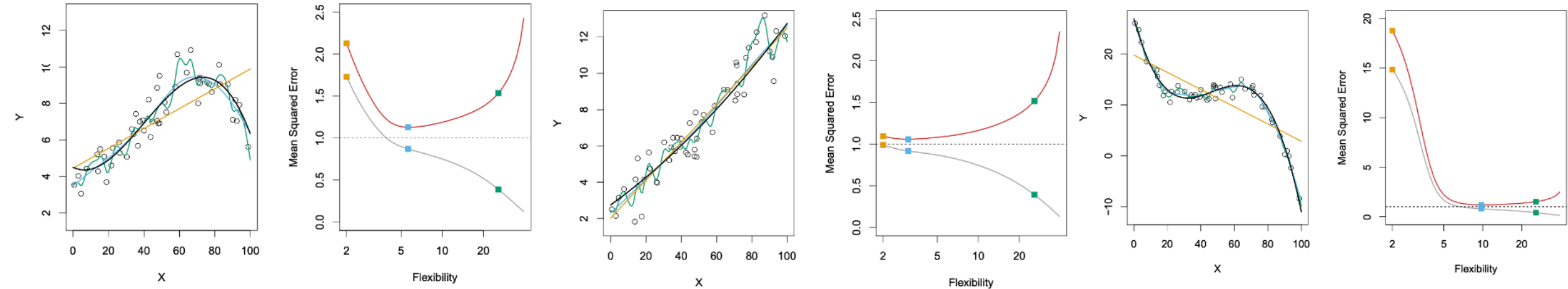
$$E \left(y_0 - \hat{f}(x_0) \right)^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon).$$

Expected test MSE:
avg. test MSE obtained
by estimating f on a
large number of
training sets

Amount by which
 $\hat{f}$ would change if we
estimated it using a
different training set

Error introduced by
approximating a real-
life f using a much
simpler model

Irreducible error



IE203 Fall Semester 2026

$$E \left(y_0 - \hat{f}(x_0) \right)^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon).$$



How To Avoid Overfitting?

Cross-validation

Using part of the training data to estimate the test MSE

...and other tricks that we will discover along the way



What About Classification Problems?

Instead of MSE, we can use the (training) error rate:

$$\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

Again, we are interested in the test error rate:

$$Avg(I(y_0 \neq \hat{y}_0))$$



Theroretical Result: Bayes Classifier

The test error is minimized by a very simple classifier that assigns each observation to the most likely class, given its predictor values

That is, we should simply assign the observation with predictor x_0 to the class j for which the following probability is largest:

$$Pr(Y = j|X = x_0)$$

The error made by the Bayes classifier is only the *irreducible* error:

$$1 - E(\max_j Pr(Y = j|X))$$



Why the Bayes Classifier?

$$1 - E(\max_j \Pr(Y = j|X))$$

Usually, we only have observations X , but we do not know the distributions that generates them

So, the Bayes classifier remains an *unattainable* gold standard

When we simulate data from a given distribution, then we can also calculate this *gold standard*



KNN Classifier (K-Nearest Neighbors)

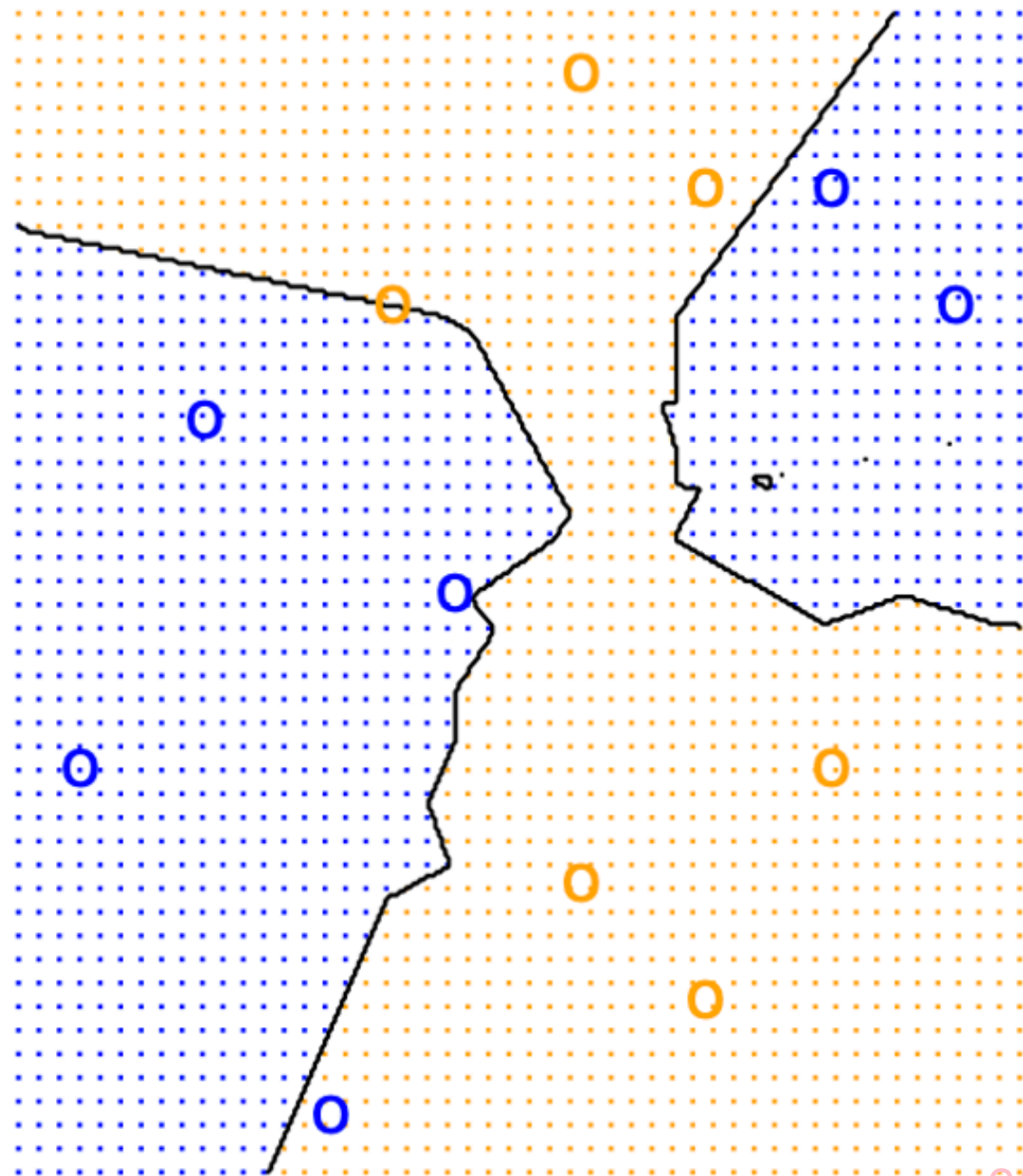
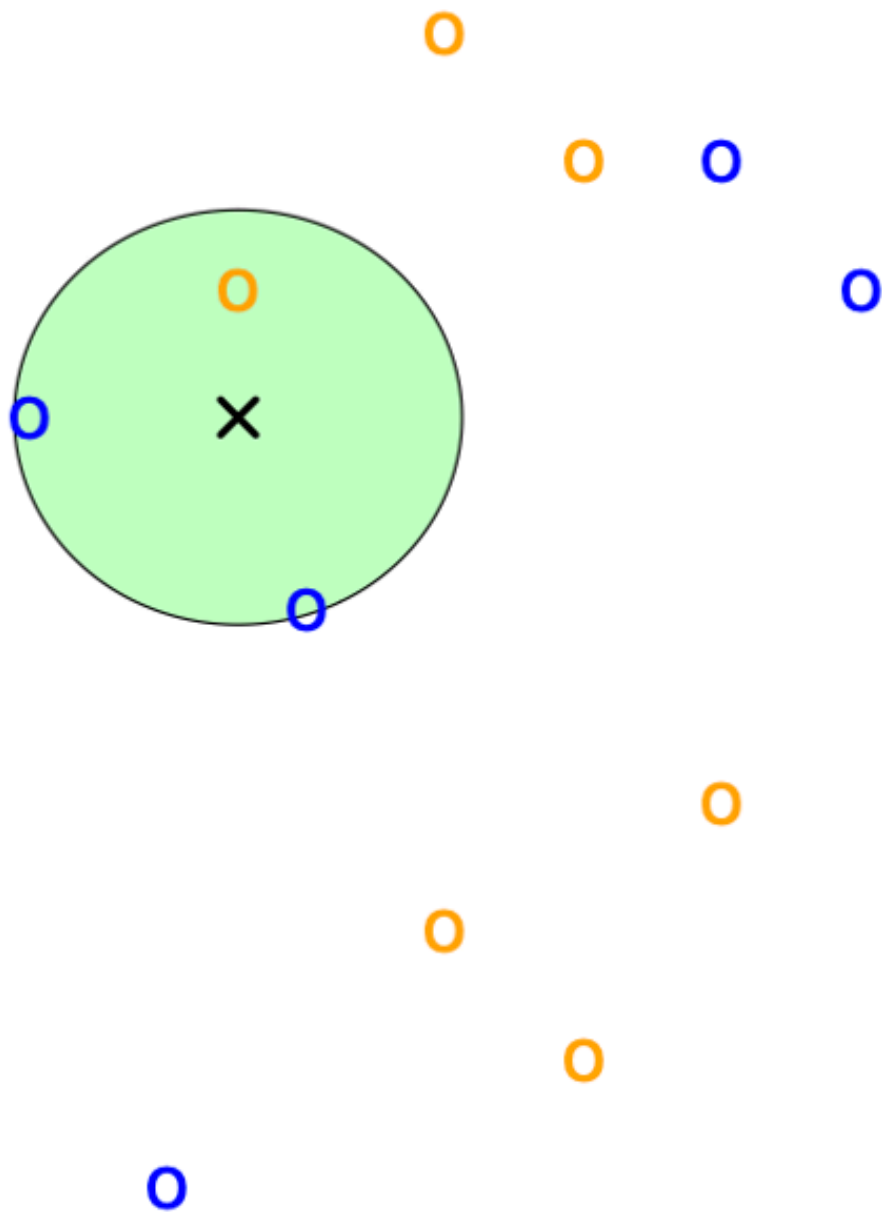
We do not know the conditional distribution of Y given X , but we can estimate it 😊, and then used the Bayes classifier using the estimate

Given a test observation x_0 , the KNN classifier first identifies the K points ($\mathcal{N}_0$) in the training data set closest to x_0

Then, estimate the conditional probability for class j as the fraction of points in ($\mathcal{N}_0$) with response equal to j

The class j associated with the largest conditional probability is chosen to classify x_0

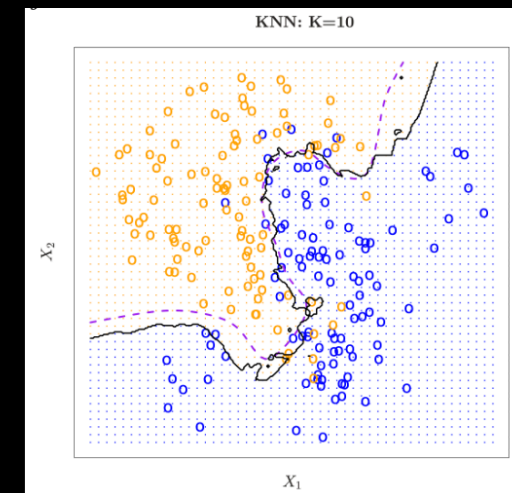
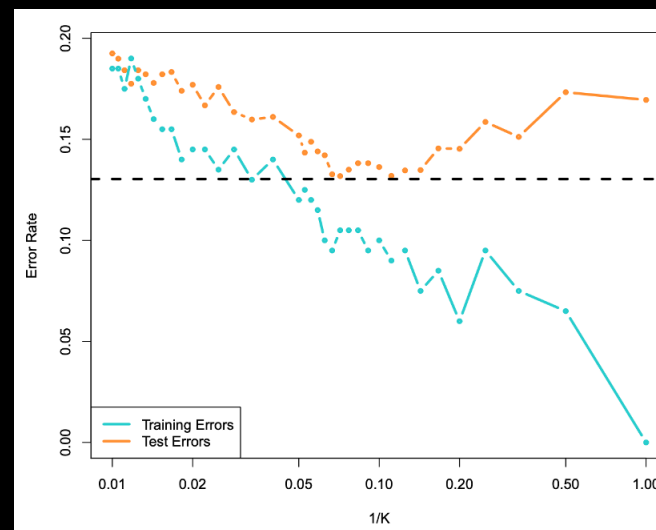
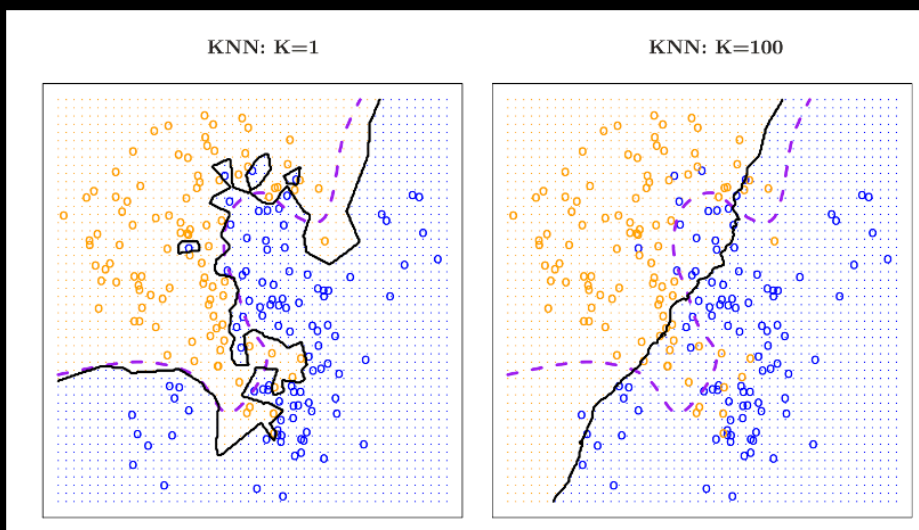
$$Pr(Y = j|X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$





KNN v. Bayes Decision Boundaries

Lower $K \longrightarrow$ More Flexible Model





What Did We Learn?

What is data mining?

Prediction v. Inference

Parametric v. Non-parametric models and model flexibility

Supervised v. Unsupervised, Regression v. Classification

How to evaluate the quality of model fit

Training v. Test error, bias-variance trade-off, overfitting

Bayes and KNN classifiers

